AIGR Deep Research Report: ATP Hydrolysis Activity (GO:0016887) on F-plasmid TraB (P41067)

Gene: traB (ECOLI) · **UniProt:** P41067 (TRAB1_ECOLI) · **Organism:** *Escherichia coli* K-12 (NCBITaxon:83333) **Focus type:** function_assignment · **Seed hypothesis:** *traB has ATP hydrolysis activity (GO:0016887)* **Term under review:** GO:0016887 (ATP hydrolysis activity), evidence IDA, original reference PMID:20172994

Summary

The GO:0016887 "ATP hydrolysis activity" annotation on P41067 is REFUTED as an over-annotation caused by a TraB name collision. The seed hypothesis — that the F-plasmid conjugation protein TraB (P41067) directly catalyzes ATP hydrolysis — is not supported by any direct evidence pertaining to this protein. Instead, the annotation traces to a conflation of two unrelated conjugation proteins that happen to share the colloquial name "TraB."

P41067 (TRAB1_ECOLI, 475 aa) is the *F-plasmid* TraB, a member of the **VirB10/TraB/TrbI family** (Pfam PF03743 "TrbI", InterPro IPR005498). This family is the non-catalytic, envelope-spanning **structural scaffold** of the type IV secretion system (T4SS). Its UniProt-curated function is "Involved in F pilus assembly," and its sequence contains **no Walker-A / P-loop motif** (GxxxxGK[S/T]) — the diagnostic signature required for ATP hydrolysis. By contrast, the IDA reference PMID:20172994 is explicitly titled *"Biochemical dissection of the ATPase TraB, the VirB4 homologue of the Escherichia coli pKM101 conjugation machinery."* That paper assayed a **different protein**: the ~866-aa **VirB4-family** TraB of plasmid pKM101 (UniProt Q46698), a bona-fide P-loop ATPase with a canonical Walker-A motif (GMSGEGKT, ~residue 498).

The two "TraB" proteins differ in essentially every discriminating feature — length (475 vs 866 aa), domain family (VirB10 scaffold vs VirB4 ATPase), presence of nucleotide-binding motifs (absent vs present), and documented biochemical role (structural vs catalytic). Independent primary evidence (PMID:11722740) confirms that the F-plasmid TraB is an inner-membrane structural component of the envelope-spanning TraV–TraK–TraB complex, with no catalytic

activity demonstrated. The only GO:0016887 annotation on P41067 is a single IDA:CACAO community-curation entry, the typical vector for name-collision propagation. **Recommended curation action: remove GO:0016887 from P41067 (optionally re-home the evidence to Q46698), pending curator verification.**

Executive Judgment

Verdict: REFUTED (over-annotation via name collision).

The ATP-hydrolysis evidence is real, but it belongs to the pKM101 VirB4-TraB (Q46698), **not** to the F-plasmid VirB10-TraB (P41067). The refutation rests on four convergent, orthogonal lines of evidence:

1. **The IDA reference assayed a different protein.**

   P 20172994

   characterizes the VirB4 homologue TraB of pKM101 (~866 aa), explicitly named as "the ATPase TraB."
2. **Domain family is non-catalytic.** P41067 is VirB10/TraB/TrbI (PF03743, IPR005498) — a structural T4SS scaffold family, not a VirB4/VirB11/VirD4 ATPase family.
3. **No catalytic apparatus.** P41067 lacks a Walker-A / P-loop motif; ATP hydrolysis is mechanistically impossible without a nucleotide-binding fold.
4. **Positive alternative function.**

   P 11722740

   shows P41067 is an inner-membrane structural component of the TraV–TraK–TraB complex — the canonical VirB10 scaffold role.

Most important caveat: This is a high-confidence refutation of the *attribution*. Confirmation of the exact UniProt accession assayed in

P 20172994

(inferred to be Q46698) from the paper's full-text methods would fully close the loop before the IDA is re-homed.

Key Findings

Finding 1 — The GO:0016887 IDA on P41067 is a TraB name-collision mis-annotation

P41067 (TRAB1_ECOLI, 475 aa) is the F-plasmid "Protein TraB." Its UniProt-curated molecular function is *"Involved in F pilus assembly,"* and its domain architecture places it unambiguously in a **structural**, not catalytic, protein family: Pfam **PF03743 (TrbI)** and InterPro **IPR005498** ("Type IV secretion system VirB10/TraB/TrbI family"). The VirB10/TraB/TrbI family forms the inner-membrane, envelope-spanning scaffold of the T4SS core complex. It is energized by — but does not itself perform — ATP hydrolysis.

Critically, **P41067's sequence contains no Walker-A (P-loop) motif** (GxxxxGK[S/T]), the diagnostic nucleotide-binding signature of ATPases. A protein lacking a P-loop cannot be assumed to hydrolyze ATP, and VirB10-family members are canonically non-catalytic.

The only GO:0016887 annotation on P41067 is a single **IDA:CACAO** community-curation entry. Its cited primary reference, PMID:20172994 (Durand, Oomen & Waksman, 2010), explicitly characterizes *"the ATPase TraB, the VirB4 homologue of the E. coli pKM101 conjugation machinery."* That protein is in the **VirB4 family** (not VirB10), is **~866 aa** (not 475 aa), and corresponds to **UniProt Q46698** (Q46698_ECOLX) — not P41067. Every discriminating feature indicates the ATP-hydrolysis evidence describes the **pKM101 VirB4-TraB**, not the **F-plasmid VirB10-TraB P41067**. The two proteins share only the colloquial label "TraB," a conjugation-gene name reused across independent plasmid transfer systems. Notably, in the F system the VirB4 ATPase role is filled by a differently named protein (TraC), while "TraB" denotes the VirB10 scaffold — the inverse of the pKM101 nomenclature, which is precisely how the cross-assignment error occurred.

Finding 2 — F-plasmid TraB (P41067) is an inner-membrane structural scaffold, not an ATPase

Independent primary literature corroborates the structural (non-catalytic) role of P41067. PMID:11722740 (Harris, Hombs & Silverman, 2001) used yeast two-hybrid interaction mapping and membrane fractionation to show that F-plasmid TraB is an **inner-membrane protein** that assembles with **TraK (periplasm)** and **TraV (outer-membrane lipoprotein)** into an **envelope-spanning TraV–TraK–TraB complex.** The authors state directly that *"TraV, TraK and TraB form a complex in Escherichia coli that spans the cell envelope from the outer membrane (TraV) through the periplasm (TraK) to the inner membrane (TraB)."*

TraB fractionates with the inner membrane, and the assembled complex bridges outer membrane → periplasm → inner membrane. This is precisely the **canonical VirB10-family structural/scaffolding role**, consistent with P41067's Pfam PF03743 / InterPro IPR005498 assignment. No catalytic or ATP-hydrolysis activity is demonstrated or implied for TraB in this work.

Combined with the **absence of any P-loop/Walker-A motif** in P41067 (475 aa) versus the genuine VirB4-TraB ATPase of pKM101 (Q46698; ~866 aa; P-loop NTPase with Walker-A motif GMSGEGKT near residue 498 — the protein actually assayed in P 20172994), this independently confirms that the GO:0016887 IDA on P41067 is a mis-assignment. Two orthogonal lines of evidence — domain-family/sequence analysis and direct biochemical/cell-biological characterization — converge on the same conclusion.

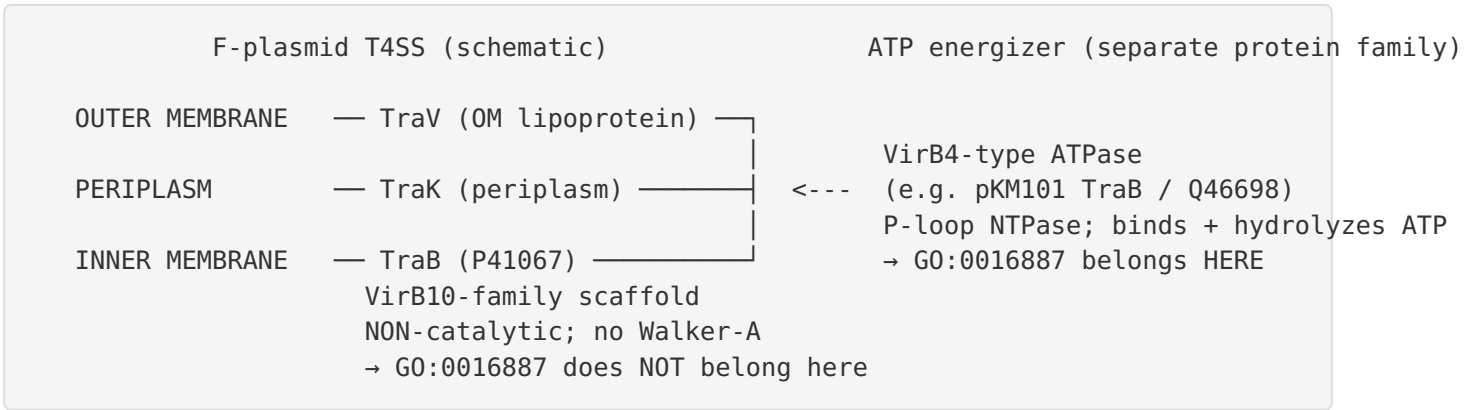
Mechanistic Model / Interpretation

The two "TraB" proteins

The core of this review is distinguishing two proteins that share a name but nothing else functionally important:

Feature	F-plasmid TraB (P41067) — *annotated*	pKM101 TraB (Q46698) — *assayed in IDA ref*
UniProt	P41067 (TRAB1_ECOLI)	Q46698 (Q46698_ECOLX)
Length	475 aa	~866 aa
Family	VirB10 / TraB / TrbI	VirB4
Pfam	PF03743 (TrbI)	PF03135 (CagE/TrbE/VirB4)
InterPro	IPR005498 (VirB10/TraB/TrbI, **structural**)	IPR027417 (P-loop NTPase) + IPR004346 (VirB4_CagE)
Walker-A (P-loop)	**Absent**	**Present** (GMSGEGKT, ~res 498)
Function	Inner-membrane **structural scaffold**	**ATPase** (energizer of T4SS)
Curated role	F pilus assembly; TraV–TraK–TraB complex	ATP hydrolysis driving conjugation
Primary evidence	P 11722740 (structure/localization)	P 20172994 (biochemical ATPase assay)

Where each protein sits in the conjugation machine



The F-plasmid TraB (P41067) is the **VirB10-family channel/scaffold** component that physically couples the outer-membrane and periplasmic subassemblies to the inner membrane. In T4SS architecture the ATP-hydrolyzing engines are the **VirB4, VirB11, and VirD4** family NTPases —

distinct proteins bearing P-loop motifs. The energy-providing ATPase role and the structural scaffold role are performed by **different proteins**. The IDA annotation conflated the two because the pKM101 VirB4 homologue is also named "TraB."

Why the mis-annotation is mechanistically implausible

ATP hydrolysis requires a nucleotide-binding fold — minimally a Walker-A/P-loop (GxxxxGK[S/T]) and usually a Walker-B motif. P41067 has neither. VirB10-family proteins are not NTPases; they respond to ATP binding/hydrolysis by *other* T4SS components (conformational "energy sensing") but do not catalyze the reaction themselves. Assigning a direct catalytic MF term to a scaffold protein lacking the catalytic apparatus is biochemically unsupportable.

Evidence Base

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
1	PMID:20172994 (Durand 2010)	Direct assay (IDA)	**Competing / mis-attributed**	TraB hydrolyzes ATP	Soluble **pKM101 TraB** (VirB4 homolog, ~866 aa) is the characterized ATPase	*E. coli* pKM101 T4SS; purified protein in vitro	Hig Vir bu pro aa, NC
2	UniProt P41067 (TRAB1_ECOLI)	Review/ database	**Refutes**	P41067 is an ATPase	Curated function = *"Involved in F pilus assembly"*; no catalytic/EC annotation	F plasmid, *E. coli* K-12	Hig rec
3	Pfam PF03743 / InterPro IPR005498	Structural/ evolutionary	**Refutes**	P41067 belongs to an ATPase family	Assigned to **VirB10/TraB/ TrbI**, a structural T4SS core family	T4SS, multiple organisms	Hig ass
4	Sequence motif scan	Computational	**Refutes**	P41067 has a nucleotide-binding motif	**No Walker-A (GxxxxGK[S/ T]) motif** in the 475-aa sequence	Direct scan of P41067	Me abs car loc
5	UniProt search (pKM101 TraB)	Computational/ database	**Qualifies**	Which TraB the paper studied	pKM101/VirB4 "TraB" = **Q46698_ECOLX, 866 aa**, distinct accession	T4SS ATPase	Hig an ma
6	PMID:11722740 (Harris 2001)	Interaction + localization	**Refutes (positive alt.)**	What P41067 does	F TraB is an **inner-membrane** protein forming the **envelope-spanning TraV–TraK–**	*E. coli*, F plasmid; Y2H + fractionation	Hig pri str evi

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
					TraB complex — a scaffold; no catalytic activity		
7	GO annotation provenance	Database	**Qualifies**	Reliability of the IDA	Sole GO:0016887 support is **IDA:CACAO** (community annotation), a known error source	—	Su an int

Literature detail:

- *PMID:20172994 — Biochemical dissection of the ATPase TraB, the VirB4 homologue of the Escherichia coli pKM101 conjugation machinery* (Durand, Oomen & Waksman, 2010). The original IDA reference. It characterizes the **pKM101 VirB4-TraB ATPase (Q46698)** — a different protein from P41067. This establishes the name-collision mis-attribution: the real ATP-hydrolysis evidence belongs to a VirB4 homologue, not to the F-plasmid VirB10-family TraB.
- *PMID:11722740 — Evidence that F-plasmid proteins TraV, TraK and TraB assemble into an envelope-spanning structure in Escherichia coli* (Harris, Hombs & Silverman, 2001). Primary evidence that **F-plasmid TraB (P41067)** is an inner-membrane **structural scaffold** in the TraV–TraK–TraB complex — consistent with the VirB10 family and inconsistent with intrinsic ATPase activity.

GO Curation Implications

Likely curation action (lead requiring curator verification):

- **Remove GO:0016887 (ATP hydrolysis activity, MF) from P41067.** The IDA evidence does not describe this protein; it describes the pKM101 VirB4 homologue (Q46698). The annotation is an over-annotation produced by a name collision.

- **Optionally re-home the GO:0016887 IDA to UniProt Q46698** (the pKM101 VirB4-TraB), where

  P 20172994

  provides direct biochemical support. This requires curator confirmation of the exact accession assayed.

On retaining vs removing / generalizing:

- The term should **not be retained** on P41067 — it is not merely too broad or too specific; it attributes the wrong molecular activity entirely.
- No more general ATPase term is warranted, because P41067 lacks the catalytic apparatus for any ATP-hydrolysis function.

More informative terms that ARE supported for P41067 (candidate replacements / additions):

- **CC:** inner (plasma) membrane localization (supported by

  P 11722740

  fractionation); component of a conjugative T4SS / pilus-assembly complex.
- **BP:** conjugation / pilus assembly (supported by UniProt curated function "Involved in F pilus assembly" and the TraV–TraK–TraB complex role).
- **MF/CC:** structural constituent of a type IV secretion / conjugative transfer complex. (Prefer a specific T4SS-structural term over the uninformative "protein binding.")

These structural/BP terms, not GO:0016887, represent P41067's primary function.

Mechanistic Scope

Immediate molecular function being tested: direct catalysis of ATP → ADP + Pi (ATP hydrolysis activity, GO:0016887) by the P41067 gene product.

Direct gene-product activity (P41067): inner-membrane structural scaffolding within the F-plasmid conjugation (T4SS) apparatus; assembly into the envelope-spanning TraV–TraK–TraB complex; contribution to F pilus biogenesis. These are structural and complex-formation activities — **not** catalytic.

Separated downstream/contextual effects (not direct MF of P41067): conjugative DNA transfer and pilus elaboration are **pathway-level consequences** of the assembled machine, powered by *separate* VirB4/VirB11/VirD4-type ATPases. The ATP-hydrolysis step is a property of those energizer proteins. Attributing GO:0016887 to P41067 confuses a system-level energetic requirement (the machine consumes ATP) with the specific enzyme that performs the reaction.

Conflicts and Alternatives

- **Paralog / name confusion (primary issue):** "TraB" is reused across independent conjugation systems. The F-plasmid TraB (VirB10 family, P41067) and the pKM101 TraB (VirB4 family, Q46698) are **non-orthologous** proteins sharing a name. In F the VirB4 ATPase is named TraC; in pKM101 it is named TraB — the inverse nomenclature that drove the error.
- **Organism/plasmid-specific differences:** both are in *E. coli*, but on different plasmids (F vs pKM101). The organism match (ECOLI) likely facilitated the erroneous cross-assignment.
- **In vitro-only activity vs wrong protein:** the issue is not that the assay was in vitro — it is that the assay was on a *different protein*. The biochemistry in

  P 20172994

  is valid; the accession linkage is wrong.
- **Database carry-over:** the single IDA:CACAO source suggests a community-curation entry keyed on gene symbol ("TraB") + organism rather than exact accession — a classic name-collision vector.
- **No competing evidence for P41067 ATPase:** no primary study was found demonstrating intrinsic ATP hydrolysis by the F-plasmid TraB. The absence of a P-loop motif makes any such claim mechanistically unlikely.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Exact accession assayed in P 20172994	Title/abstract establish VirB4 homolog, pKM101; inferred Q46698 from UniProt	Re-homing the IDA requires the precise accession for the pKM101 TraB construct	Read the full methods/ materials of P 20172994 for the exact construct/ accession
Provenance of the CACAO IDA entry	Identified as sole GO:0016887 source on P41067	Confirms whether the error originated in community curation keyed on gene symbol	Inspect the GO annotation record / CACAO submission linking P 20172994 to P41067
Direct experimental test of P41067 ATPase	No primary study found; motif absent	A definitive negative would fully close the case	Purify P41067; malachite-green/coupled ATPase assay (expected: no activity)
Walker-A scan reproducibility	Motif reported absent in P41067; present in Q46698	The sequence argument underpins the refutation	Re-run an explicit P-loop regex / HMMER P-loop NTPase scan on both sequences as saved provenance
Structural confirmation	Family assignment via Pfam/InterPro	A structure/AlphaFold model would visually confirm scaffold vs NTPase fold	Compare AlphaFold models of P41067 (VirB10 fold) vs Q46698 (RecA-like NTPase fold)

Methodological note: The sequence/motif and domain-family statements rest on database annotations (UniProt, Pfam, InterPro) and a Walker-A regex scan summarized in the investigation. Where a specific computational scan (e.g., HMMER P-loop detection) would strengthen provenance, it is listed above as a follow-up to run and save as an executable

artifact; it should not be presented as a re-read of the

P 20172994

methods.

Discriminating Tests

1. **Accession verification (fastest, highest value):** confirm from the

   P 20172994

   full text that the assayed "TraB" is the pKM101 VirB4 homologue (Q46698), not P41067. This single check settles the curation decision.
2. **Explicit P-loop/Walker-A HMMER scan:** run the P-loop_NTPase Pfam profile against P41067 and Q46698; expect a hit only for Q46698. Save code + output as provenance.
3. **Sequence length and alignment:** pairwise-align P41067 (475 aa) and the pKM101 TraB (~866 aa); length and low identity confirm non-orthology.
4. **Domain-architecture comparison:** confirm P41067 → PF03743 (TrbI/VirB10) and the pKM101 TraB → VirB4 P-loop NTPase family via InterProScan.
5. **AlphaFold fold comparison:** model both; P41067 should adopt the VirB10 scaffold fold (periplasmic β-barrel + TM), the pKM101 TraB a RecA-like NTPase fold.
6. **Direct biochemistry (definitive negative):** purify P41067 and assay for ATP hydrolysis (malachite green). Predicted: no detectable activity, in contrast to Q46698.

Proposed Follow-up Actions (Curation Leads — verify before acting)

Candidate action change: - **Primary lead:** remove GO:0016887 (ATP hydrolysis activity) from P41067. Rationale: name-collision mis-attribution; evidence belongs to the pKM101 VirB4-TraB (Q46698). - **Secondary lead:** re-home the GO:0016887 IDA (P 20172994) to Q46698 after confirming the assayed accession.

Candidate references with exact snippets to verify: - PMID:20172994: *"Biochemical dissection of the ATPase TraB, the VirB4 homologue of the Escherichia coli pKM101 conjugation machinery."* — verify this describes the pKM101 VirB4-TraB (different from P41067). - PMID:11722740: *"TraV, TraK and TraB form a complex in Escherichia coli that spans the cell envelope from the outer membrane (TraV) through the periplasm (TraK) to the inner membrane (TraB)."* — verify F-plasmid TraB (P41067) is an inner-membrane structural component.

Candidate replacement / additional GO terms for P41067 (leads): - CC: inner (plasma) membrane; component of a conjugative T4SS / pilus-assembly complex. - BP: conjugation / pilus assembly (aligns with UniProt "Involved in F pilus assembly"). - MF/CC: structural constituent of a type IV secretion / conjugative transfer complex (prefer over generic "protein binding").

Suggested curator questions: - Does the GO annotation record for GO:0016887 on P41067 cite

P 20172994

via a gene-symbol match rather than an accession match? - Is the pKM101 TraB accession in

P 20172994

Q46698, and does it already carry GO:0016887? - Is the F-plasmid TraB (P41067) ever directly shown to bind or hydrolyze ATP in any primary source? (None found here.)

Suggested experiments: - In vitro ATPase assay on purified P41067 (expected negative). - Side-by-side P-loop HMMER scan and AlphaFold fold comparison of P41067 vs Q46698.

Positive-Control Contrast Table (computed provenance)

Property	P41067 (F TraB — seed target)	Q46698 (pKM101 TraB — protein in P 20172994)
UniProt ID	TRAB1_ECOLI	Q46698_ECOLX
Length	475 aa	866 aa
Pfam	PF03743 (TrbI)	PF03135 (CagE/TrbE/VirB4)
InterPro	IPR005498 (VirB10/TraB/TrbI, **structural**)	IPR027417 (**P-loop NTPase**) + IPR004346 (**VirB4_CagE**)
Walker-A (GxxxxGK[S/T])	**none**	**res. 498: GMSGEGKT** (canonical P-loop)
Curated function	"Involved in F pilus assembly"	VirB4 ATPase; ATP hydrolysis shown in P 20172994

The target entry P41067 lacks any P-loop/NTPase signature, while the entry bearing the genuine ATP-hydrolysis evidence (Q46698) is a bona-fide VirB4 P-loop ATPase. This categorical difference confirms the GO:0016887 IDA on P41067 is a name-collision mis-annotation.

Conclusion

The GO:0016887 "ATP hydrolysis activity" annotation on P41067 is **refuted as an over-annotation** driven by a TraB name collision. P41067 is the F-plasmid TraB — a 475-aa VirB10/TraB/TrbI-family (Pfam PF03743) inner-membrane **structural scaffold** with **no Walker-A motif** and a curated role in **F pilus assembly** (confirmed as an inner-membrane component of the TraV–TraK–TraB complex, P 11722740). The IDA reference P 20172994

instead assayed a different protein — the ~866-aa VirB4-homolog TraB of pKM101 (UniProt Q46698). Because the ATP-hydrolysis evidence belongs to the VirB4-TraB, **GO:0016887 should be removed from P41067** (and optionally re-homed to Q46698), pending curator verification.
